

Practical 1 — Community ecology of freshwater fishes

Master FBE · Functional Ecology in R

In this practical you will analyse species richness, community composition and dark diversity across river basins worldwide using a subset of the GlobalTraits / FishMORPH databases. All data are provided in `fish_communities.csv` and `env_data.csv`.

Load the data:

```
library(vegan); library(ggplot2); library(dplyr)

# Community matrix: sites (river basins) × species
comm <- read.csv("fish_communities.csv", row.names=1)

# Environmental variables per basin
env <- read.csv("env_data.csv", row.names=1)

# Quick look
head(comm[, 1:5], 3)
head(env)
```

Question 1 — Species richness (15 pts)

1a. Compute species richness S per basin and add it to the `env` data frame. Which biogeographic realm has the highest mean richness? Display a ranked barplot coloured by realm.

```
S <- specnumber(comm)
env$S <- S

# Mean richness per realm
env |> group_by(Realm) |> summarise(mean_S=mean(S)) |> arrange(desc(mean_S))

# Barplot
ggplot(env,aes(reorder(Basin,S),S,fill=Realm)) +
  geom_col() + coord_flip() + theme_bw() +
  labs(x="Basin",y="Species richness S")
```

1b. Rarefy all basins to the minimum sample size. Does the ranking change? Explain why rarefaction matters.

```
min_n <- min(rowSums(comm))
S_rare <- rarefy(comm,min_n)
env$S_rare <- S_rare
cor.test(env$S, env$S_rare, method="spearman")
```

→ Write your interpretation (3-4 sentences) in the space below or in a separate document:

Question 2 — Beta diversity (20 pts)

2a. Compute Bray-Curtis dissimilarity. Run an NMDS (k=2) and plot basins coloured by realm. Do realms form distinct clusters?

```
d_bray <- vegdist(comm,"bray")
ord <- metaMDS(d_bray,k=2,trymax=50)
cat("Stress:",round(ord$stress,3))

sc <- as.data.frame(scores(ord,"sites"))
sc$Realm <- env$Realm
ggplot(sc,aes(NMDS1,NMDS2,colour=Realm))+geom_point(size=3)+
  stat_ellipse(linewidth=0.7)+theme_bw()
```

2b. Decompose beta diversity into **replacement** and **nestedness** components using betapart. Which component dominates? What does this mean ecologically?

```
library(betapart)
dune_pa <- decostand(comm,"pa")
beta_pair <- beta.pair(dune_pa)
hist(as.vector(beta_pair$beta.sim),breaks=20,
     main="Replacement component",col="#27ae60")
```

2c. Test whether realm explains community composition using PERMANOVA (adonis2).

```
adonis2(comm~Realm,data=env,permutations=999,by="margin")
```

Question 3 — Dark diversity (20 pts)

3a. Estimate dark diversity for each basin using Beals smoothing (threshold = 0.3). Compute completeness = $S / (S + DD)$.

```
beals_scores <- beals(comm)
dark <- (comm==0) & (beals_scores>0.3)
DD <- rowSums(dark)
env$DD <- DD
env$completeness <- env$S/(env$S+DD)
```

3b. Plot completeness against S. Are species-rich basins more or less complete? Test the relationship with a linear model and interpret the slope.

```
ggplot(env,aes(S,completeness,colour=Realm))+
  geom_point(size=2.5)+geom_smooth(method="lm")+
  labs(x="Observed richness",y="Completeness")+theme_bw()

m <- lm(completeness~S,data=env)
summary(m)
```

3c. Which realm has the lowest completeness? Propose two ecological explanations.